

## **Module-3 : Hardware and Components Basics Understanding of Hardware and its components**

### Section 1: Multiple Choice

1. What is the primary function of a router in a computer network?
  - a) Assigning IP addresses to devices
  - b) Providing wireless connectivity to devices
  - c) Forwarding data packets between networks
  - d) Managing user authentication and access control

ANS: c) ) Forwarding data packets between networks

2. What is the purpose of DNS (Domain Name System) in a computer network?
  - a) Encrypting data transmissions for security
  - b) Assigning IP addresses to devices dynamically
  - c) Converting domain names to IP addresses
  - d) Routing data packets between network segments

ANS: c) Converting domain names to IP addresses

3. What type of network topology uses a centralized hub or switch to connect all devices?
  - a) Star
  - b) Bus
  - c) Ring
  - d) Mesh

ANS: a) Start

4. Which network protocol is commonly used for securely accessing and transferring files over a network?
  - a) HTTP
  - b) FTP
  - c) SMTP
  - d) POP3

ANS: b) FTP

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### Section 2 : True or False

5. True or False: A firewall is a hardware or software-based security system that monitors and controls incoming and outgoing network traffic based on predetermined security rules.

ANS: TRUE

6. True or False: DHCP (Dynamic Host Configuration Protocol) assigns static IP addresses to network devices automatically.

ANS: FALSE

7. True or False: VLANs (Virtual Local Area Networks) enable network segmentation by dividing a single physical network into multiple logical networks.

ANS: TRUE

### Section 3 : Short Answer

8. Explain the difference between a hub and a switch in a computer network.

ANS:

#### Hub

A hub broadcasts data to all devices connected to it.

Works at the Physical Layer (Layer 1) of the OSI model.

Less efficient because all devices receive the same data.

Causes more network traffic and collisions.

Slower and less secure.

Older technology and rarely used today.

#### Switch

A switch sends data only to the specific device it is meant for.

Works at the Data Link Layer (Layer 2) of the OSI model.

More efficient because data goes directly to the destination device.

Reduces network traffic and collisions.

Faster and more secure.

Commonly used in modern networks.

9. Describe the process of troubleshooting network connectivity issues.

ANS:

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### **A. Identify the Problem**

Determine whether the issue affects one device or the entire network.

### **B. Check Physical Connections**

Ensure cables, router, modem, and Wi-Fi connections are working properly.

### **C. Restart Network Devices**

Reboot the computer, router, and modem to fix temporary issues.

### **D. Verify IP Configuration**

Check that the device has a valid IP address from DHCP.

### **E. Test Connectivity**

Use tools like ping to check communication between devices and the internet.

### **F. Check Firewall and Network Settings**

Ensure firewall, DNS, and network settings are not blocking the connection.

## **Section 4 : Practical Application**

10. Demonstrate how to configure a wireless router's security settings to enhance network security.

ANS:

### **Configuring Wireless Router Security Settings**

1. Log in to the Router
  - Open a web browser and enter the router's IP address (such as 192.168.1.1).
  - Enter the administrator username and password.
2. Change the Default Admin Password
  - Replace the default router password with a strong, unique password.
3. Set Strong Wi-Fi Encryption
  - Go to Wireless Security settings.
  - Select WPA2 or WPA3 security mode for better protection.
4. Create a Strong Wi-Fi Password
  - Use a password with letters, numbers, and special characters.
  - Avoid easy passwords like "12345678".
5. Change the Default Network Name (SSID)
  - Rename the Wi-Fi network to something unique that does not reveal personal information.

### **Module-3 : Hardware and Components Basics Understanding of Hardware and its components**

6. Disable WPS (Wi-Fi Protected Setup)
  - Turn off WPS because it can make the network vulnerable to attacks.
7. Update Router Firmware
  - Check for firmware updates regularly to fix security vulnerabilities.
8. Save and Restart
  - Save all settings and restart the router to apply the changes.

#### **Section 5 : Essav**

11. Discuss the importance of network documentation and provide examples of information that should be documented.

ANS :

#### **Importance of Network Documentation**

Network documentation is important because it helps organizations manage, maintain, and troubleshoot their networks efficiently. Proper documentation improves security, reduces downtime, and makes it easier for technicians to understand the network setup.

#### **Importance**

1. Helps in troubleshooting network problems quickly.
2. Makes network maintenance and upgrades easier.
3. Improves network security and management.
4. Reduces downtime during failures.
5. Assists new IT staff in understanding the network.
6. Provides a backup reference for future use.

#### **Examples of Information to Document**

- Network topology diagrams
- IP addresses and subnet details
- Router, switch, and firewall configurations
- Device names and locations
- Wi-Fi SSIDs and security settings
- User accounts and access permissions
- Cable connections and port numbers
- Internet Service Provider (ISP) details and contact information